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## This is the free Material Data Center Datasheet of Ultramid® B3S - PA6 - BASF

Material Data Center offers the following functions for **Ultramid® B3S**:

unit conversion, PDF datasheet print, comparison with other plastics, application examples, snap fit calculation, beam deflection calculation, parameters for material models, CAE Interfaces

Check here, which other **Ultramid B** datasheets, application examples or technical articles are available in Material Data Center

Use the following short links to get directly to the properties of interest in this datasheet:

### Product Texts

An easy flowing, finely crystalline injection moulding grade for very fast processing. Parts produced include thin-walled technical parts (eg housing, fittings, grips, small parts and fixing clamps).

| Processing/Physical Characteristics         | dry / cond        | Unit                   | Test Standard        |
|---------------------------------------------|-------------------|------------------------|----------------------|
| <b>ISO Data</b>                             |                   |                        |                      |
| Melt volume-flow rate, MVR                  | 160 / *           | cm <sup>3</sup> /10min | ISO 1133             |
| Temperature                                 | 275 / *           | °C                     | -                    |
| Load                                        | 5 / *             | kg                     | -                    |
| Molding shrinkage, parallel                 | 0.9 / *           | %                      | ISO 294-4, 2577      |
| Molding shrinkage, normal                   | 0.9 / *           | %                      | ISO 294-4, 2577      |
| Ejection temperature                        | 145               | °C                     | -                    |
| <b>Mechanical properties</b>                | <b>dry / cond</b> | <b>Unit</b>            | <b>Test Standard</b> |
| <b>ISO Data</b>                             |                   |                        |                      |
| Tensile Modulus                             | 3500 / 1200       | MPa                    | ISO 527              |
| Yield stress                                | 90 / 45           | MPa                    | ISO 527              |
| Yield strain                                | 4 / 20            | %                      | ISO 527              |
| Nominal strain at break                     | 10 / >50          | %                      | ISO 527              |
| Tensile creep modulus, 1h                   | * / 1300          | MPa                    | ISO 899-1            |
| Tensile creep modulus, 1000h                | * / 1100          | MPa                    | ISO 899-1            |
| Charpy impact strength, +23°C               | 250 / N           | kJ/m <sup>2</sup>      | ISO 179/1eU          |
| Charpy impact strength, -30°C               | 200 / -           | kJ/m <sup>2</sup>      | ISO 179/1eU          |
| Charpy notched impact strength, +23°C       | 4 / 50            | kJ/m <sup>2</sup>      | ISO 179/1eA          |
| Charpy notched impact strength, -30°C       | 3 / -             | kJ/m <sup>2</sup>      | ISO 179/1eA          |
| <b>Thermal properties</b>                   | <b>dry / cond</b> | <b>Unit</b>            | <b>Test Standard</b> |
| <b>ISO Data</b>                             |                   |                        |                      |
| Melting temperature, 10°C/min               | 220 / *           | °C                     | ISO 11357-1/-3       |
| Glass transition temperature, 10°C/min      | 60 / *            | °C                     | ISO 11357-1/-2       |
| Temp. of deflection under load, 1.80 MPa    | 65 / *            | °C                     | ISO 75-1/-2          |
| Temp. of deflection under load, 0.45 MPa    | 180 / *           | °C                     | ISO 75-1/-2          |
| Vicat softening temperature, B              | 204 / *           | °C                     | ISO 306              |
| Coeff. of linear therm. expansion, parallel | 102 / *           | E-6/K                  | ISO 11359-1/-2       |
| Oxygen index                                | 26 / *            | %                      | ISO 4589-1/-2        |
| <b>Electrical properties</b>                | <b>dry / cond</b> | <b>Unit</b>            | <b>Test Standard</b> |
| <b>ISO Data</b>                             |                   |                        |                      |
| Relative permittivity, 100Hz                | 4.1 / -           | -                      | IEC 62631-2-1        |
| Relative permittivity, 1MHz                 | 3.3 / 7           | -                      | IEC 62631-2-1        |
| Dissipation factor, 100Hz                   | 100 / -           | E-4                    | IEC 62631-2-1        |
| Dissipation factor, 1MHz                    | 300 / 3000        | E-4                    | IEC 62631-2-1        |
| Volume resistivity                          | 1E13 / 1E10       | Ohm*m                  | IEC 62631-3-1        |
| Surface resistivity                         | * / 1E10          | Ohm                    | IEC 62631-3-2        |
| Electric strength                           | 40 / 29           | kV/mm                  | IEC 60243-1          |
| Comparative tracking index                  | - / 600           | -                      | IEC 60112            |
| <b>Other properties</b>                     | <b>dry / cond</b> | <b>Unit</b>            | <b>Test Standard</b> |
| Water absorption                            | 9.5 / *           | %                      | Sim. to ISO 62       |
| Humidity absorption                         | 3 / *             | %                      | Sim. to ISO 62       |
| Density                                     | 1130 / -          | kg/m <sup>3</sup>      | ISO 1183             |

| Material specific properties                | dry / cond | Unit               | Test Standard       |
|---------------------------------------------|------------|--------------------|---------------------|
| <b>ISO Data</b>                             |            |                    |                     |
| Viscosity number                            | 145 / *    | cm <sup>3</sup> /g | ISO 307, 1157, 1628 |
| Test specimen production                    | Value      | Unit               | Test Standard       |
| <b>ISO Data</b>                             |            |                    |                     |
| Injection Molding, melt temperature         | 250        | °C                 | ISO 294             |
| Injection Molding, mold temperature         | 80         | °C                 | ISO 294             |
| Injection Molding, injection velocity       | 200        | mm/s               | ISO 294             |
| Processing Recommendation Injection Molding | Value      | Unit               | Test Standard       |
| Pre-drying - Temperature                    | 80         | °C                 | -                   |
| Pre-drying - Time                           | 4          | h                  | -                   |
| Processing humidity                         | ≤0.15      | %                  | -                   |
| Melt temperature                            | 250 - 270  | °C                 | -                   |
| Mold temperature                            | 40 - 60    | °C                 | -                   |

#### Characteristics

#### Processing

Injection Molding

#### Delivery form

Pellets

#### Additives

Lubricants, Release agent

#### Regional Availability

North America, Europe, Asia Pacific, South and Central America, Near East/Africa

#### Other text information

#### Injection molding

##### PREPROCESSING

Pre/Post-processing, max. allowed water content: .15 %

Pre/Post-processing, Pre-drying, Temperature: 80 °C

Pre/Post-processing, Pre-drying, Time: 4 h

##### PROCESSING

injection molding, Melt temperature, range: 250 - 270 °C

injection molding, Melt temperature, recommended: 260 °C

injection molding, Mold temperature, range: 40 - 60 °C

injection molding, Mold temperature, recommended: 60 °C

injection molding, Dwell time, thermoplastics: 10 min

#### Application examples

Coatrack

Handle, door

Screwless electric terminals

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